

* Why cosine is preferred over Euclidean?

→ cosine similarity cares about Direction (meaning)
while Euclidean distance is affected by length (magnitude)

Euclidean Distance

$$d(A, B) = \|A - B\|$$



- * Measures absolute distance in space
- * Sensitive to vector length.

Example.

↳ Case :- Same meaning, Different magnitude.

$$A = [1, 2]$$

$$B = [2, 4] = 2A$$

∴ B is just A scaled up.

∴ same Direction

A = "I love AI"

B = "I really really love AI so much"

Euclidean → "How far?"
penalizing magnitude

$$d(A, B) = \sqrt{5}$$

∴ They are different

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cosine similarity.

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$



- * Measures angles between vectors.
- * Ignores magnitude, focuses on direction.

cosine → Are they pointing in same direction

$$B = 2A$$

$$\cos \theta = 1$$

They are identical in meaning



In real AI Systems.

- sentence length can change magnitude.
- Model layers can scale vectors.
- frequency of words affects size.

But meaning should not change.

@codingdiol

Embeddings are about direction (meaning)
not magnitude — that's why cosine wins.

Magnitude is Noisy.

- longer sentences → large vector norms.
- frequent words → scaled embedding.
- Neural layers → arbitrary scaling.

High-Dimensional problem

Embedding —

128D, 768D, 1536D